

Regime-Conditional Factor Premia and Institutional Flow Dynamics in Indian Equities

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Abstract

We document regime-conditional factor premia in Indian equity markets and examine whether daily foreign institutional investor (FII) flows explain the underlying dynamics. Using the IIM Ahmedabad survivorship-bias-adjusted factor library (Agarwalla, Jacob, and Varma, 2013) spanning 2019–2025, we identify Bull, Sideways, and Bear market states via a Hidden Markov Model and characterise factor performance within each state. The momentum premium (WML) concentrates in Sideways regimes (Sharpe ratio 1.667, $t = 3.24$), while the market premium concentrates in Bull regimes (Sharpe ratio 1.485, $t = 2.28$). Value (HML) is directionally positive in Bear regimes (Sharpe ratio 2.26), consistent with defensive rotation into low-valuation stocks, though statistical inference is limited by the sparse Bear sample ($n = 121$ days). We next test whether daily NSDL FII net flows Granger-cause individual factor premia across 15 regime-factor pairs (five daily factors $\times$ three regimes), applying Benjamini-Hochberg false discovery rate correction. No cell survives — FII aggregate flows do not predict factor premia in any regime. A buy-sell decomposition reveals that gross FII volume (both purchases and sales) predicts aggregate market returns specifically in Bear regimes ($F \approx 5$, $p \approx 0.009$), consistent with FII acting as the marginal price setter when domestic liquidity thins; because Bear observations are concentrated in the COVID episode, this result cannot be established independently of the crisis window. Realized pairwise correlation (DCC) rises significantly in Bear (mean 0.30 vs. 0.25 in Bull; $t = 22.9$), confirming diversification failure coincides with regime transitions rather than leading them. A regime-aware factor allocation strategy achieves Sharpe 0.794 versus 0.733 for buy-and-hold over the 2021–2026 walk-forward period, although a static mean-variance allocation outperforms both over this sample — a result we report transparently and discuss in Section 7.

JEL Classification: G11, G12, G15, G23 **Keywords:** factor investing, regime-conditional returns, Hidden Markov Model, Granger causality, FII flows, Indian equities, momentum, DCC correlation, institutional investors

1. Introduction

Factor investing — the systematic harvesting of return premia associated with market beta (Sharpe, 1964), size, value, momentum, profitability, and investment — is extensively documented in developed markets (Fama and French, 1993, 2015; Carhart, 1997) and across international and emerging markets (Rouwenhorst, 1998; Asness, Moskowitz, and Pedersen, 2013), but evidence for Indian equities remains limited in two respects. First, most Indian studies (e.g., Sehgal and Balakrishnan, 2013) use monthly data, precluding analysis of regime-specific dynamics through which premia are transmitted. Second, the role of foreign institutional investors (FIIs) — who account for roughly 20–25% of NSE cash market turnover — has not been examined at the factor level.

A growing practitioner literature documents that factor premia vary substantially across market regimes. Naik, Devarajan, Nowobilski, Page, and Pedersen (2016) identify regime-dependency as a first-order concern for systematic strategies: correlation structure and premia concentration shift materially between bull, sideways, and bear environments, and static factor allocations that ignore the business cycle leave substantial risk-adjusted return on the table. Systematic approaches that condition factor exposure on estimated market state have been shown to improve risk-adjusted performance relative to static factor allocations. Our paper brings this framework to the Indian market using a rigorous, returns-based regime identification approach.

The motivation for examining FII flows is both theoretical and practical. Theoretically, if FIIs trade on factor-level signals, their daily flows should carry incremental predictive power for specific factor premia beyond the factor’s own history. Practically, if the predictability concentrates in Bear regimes — when systematic strategies are most exposed and premia are most volatile — a flow-based early warning system has direct value for Indian AMCs and systematic portfolio managers. We test this hypothesis using gross flow data from the NSDL FPI archive, which allows decomposition into purchases and sales separately — a distinction most prior literature cannot make.

Our main contributions are as follows. First, we provide a comprehensive regime-conditional factor analysis for Indian equities using the canonical IIM Ahmedabad factor library at daily frequency. Second, we conduct a 15-test Granger causality analysis between FII flows and factor premia, applying Benjamini-Hochberg false discovery rate correction — a methodological standard rarely applied in Indian

market microstructure research. Third, our buy-sell decomposition reveals a regime-asymmetric market predictability finding that net-flow studies cannot uncover. Fourth, we validate all factor results against own-constructed daily factors from the NSE universe and quantify the sensitivity to factor construction methodology.

The remainder of the paper is organized as follows. Section 2 describes the data. Section 3 presents the HMM regime identification. Section 4 documents regime-conditional factor premia. Section 5 tests FII flows as factor predictors. Section 6 presents the buy-sell decomposition. Section 7 describes the regime-aware strategy. Section 8 reports robustness checks. Section 9 discusses limitations. Section 10 concludes.

2. Data

2.1 Factor Returns — Primary Dataset

Our primary factor return data are the IIM Ahmedabad Fama-French and Momentum factor series (Agarwalla, Jacob, and Varma, 2013), downloaded from <https://faculty.iima.ac.in/~iffm/Indian-Fama-French-Momentum/>. We use the daily, survivorship-bias-adjusted series (release 2025-12), which covers October 1993 through December 2025. The IIMA library provides four factors: MKT (market excess return over the risk-free rate), SMB (size), HML (value), and WML (momentum). Factors are constructed from the full BSE/NSE universe with micro-cap and penny-stock filters and point-in-time accounting data from CMIE Prowess. The IIMA construction methodology is peer-reviewed, publicly documented, and constitutes the standard benchmark for Indian factor research.

We restrict our analysis to the period January 2019–December 2025, yielding 1,669 common trading days after alignment with regime labels and FII data. Full-sample factor returns are reported in Table 1.

2.2 Factor Returns — Robustness Dataset

As a secondary dataset, we construct own daily factors from a 14-stock NSE blue-chip universe (Nifty 50 constituents selected for data completeness) using constituent price data and annual-report fundamentals sourced from yfinance, with a six-month post-fiscal-year-end lag for point-in-time compliance. This dataset provides MKT, SMB, HML, RMW (profitability), and CMA (investment) at daily frequency — enabling a five-factor comparison not available in the IIMA library. Cross-validation against

IIMA shows high correlation for MKT ($r = 0.907$), but near-zero correlation for SMB ($r = -0.18$) and HML ($r = 0.01$). We attribute this divergence to the absence of true size and value dispersion in a 14-stock large-capitalisation universe; no stock in this set qualifies as genuinely small or value relative to the full BSE universe from which IIMA constructs its long-short SMB and HML portfolios. We treat own-constructed factors as illustrative of daily factor dynamics and report regime results in Appendix A2; IIMA factors constitute the primary analysis.

Survivorship bias. Our 14-stock NSE universe includes only stocks with complete return and fundamental data over 2019–2026. Firms delisted or significantly restructured over this period — including YES Bank (capital reconstruction) and several IL&FS subsidiaries — are excluded from historical return calculations. This survivorship filter tends to inflate value and profitability premia directionally; results should be interpreted accordingly. The IIMA primary factors are survivorship-bias adjusted by construction (Agarwalla, Jacob, and Varma, 2013), which is a principal reason we use them as the primary dataset.

2.3 FII Daily Flow

We obtain daily FII net equity investment from the NSDL FPI investment activity archive (<https://www.fpi.nsdl.co.in>), spanning January 2019 to June 2026 — 1,939 trading days. Data cover gross purchases, gross sales, and net investment for equity cash markets, in ₹ crore (1 crore = ₹10 million). We normalise daily net flow by a 252-day rolling mean of absolute flow, producing a dimensionless signal comparable across time as market capitalisation grew. We verify stationarity via augmented Dickey-Fuller tests. Gross flow decomposition (purchases and sales separately) is preserved for Section 6.

2.4 Market Index and VIX

For HMM estimation and the MKT factor, we use Nifty 50 daily log returns. India VIX daily close data is sourced from NSE India (2019–2026), providing 1,734 trading days. We use log-differenced VIX for Granger tests following ADF confirmation of non-stationarity in levels.

2.5 Sample Alignment

After aligning all series on common trading days, the primary analysis covers 1,669 days (January 2019 – December 2025). Regime distribution: Bull 593 days (35.5%), Sideways 955 days (57.2%), Bear 121 days (7.3%).

3. HMM Regime Identification

Following the regime-switching framework of Hamilton (1989), we model the daily return-volatility state as a latent variable following a three-state first-order Hidden Markov process. Observable emissions are daily Nifty 50 log return, India VIX, and 20-day momentum (z-scored), with Gaussian emissions within each state. We label states post-hoc by state-conditional means: the high-return, low-VIX state is Bull; the high-VIX, low-return state is Bear; the intermediate state is Sideways.

Causal identification. We use filtered (forward-pass only) regime probabilities throughout to avoid look-ahead bias inherent in full-sample Viterbi smoothing. The filtered forward algorithm conditions the regime probability at time t only on observations through t , eliminating the retrospective smoothing that contaminates states identified by Viterbi decoding on the full sequence. Additionally, all Granger causality and factor attribution tests condition on the prior-day regime label ($t-1$), removing any simultaneity between the contemporaneous return and the state assignment. Verification checks confirm: COVID-19 crash (March 23, 2020) is correctly labeled Bear; 2021 bull market is predominantly Bull/Sideways; Bear VIX (mean 28.3) exceeds Bull VIX (mean 14.5); regime persistence diagonals exceed 0.85 for all three states.

Realized pairwise correlation from a DCC(1,1) model (Engle, 2002) estimated on GARCH(1,1)-standardized returns of 14 NSE stocks confirms the regime identification: Bear-regime average pairwise DCC correlation (mean = 0.297) is significantly higher than Bull (mean = 0.248; $t = 22.9$, $p < 0.001$), consistent with the “all correlations go to 1 in a crash” phenomenon. Notably, India VIX Granger-causes DCC average correlation ($F = 6.99$, $p < 0.001$) but not vice versa ($F = 1.05$, $p = 0.31$), confirming that the options market prices correlation stress before it appears in realized return comovement. Practitioners using realized correlation as a real-time risk trigger are therefore systematically late relative to the implied volatility signal.

4. Regime-Conditional Factor Premia

Table 1 — Full-Sample IIMA Factor Premia (2019-2025, $n = 1,669$ days)

Factor	Ann. Return %	Ann. Vol %	Sharpe	t-stat	p-value	Sig.
MKT	12.05	16.59	0.726	1.87	0.062	†
SMB	−0.39	12.40	−0.032	−0.08	0.935	
HML	10.28	13.16	0.781	2.01	0.045	*
WML	14.72	13.63	1.080	2.78	0.006	**

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$. Factors from Agarwalla, Jacob, and Varma (2013), daily frequency, survivorship-bias adjusted.

Over the full 2019-2025 sample, the market (MKT, 12.05% annualised, SR = 0.726†) and value (HML, 10.28%, SR = 0.781*) factors carry the largest risk-adjusted premia. The momentum factor (WML, 14.72%, SR = 1.080**) is the sole factor achieving significance at the 1% level in the full sample. SMB is economically small and insignificant (−0.39%, SR = −0.032).

Table 2 — Regime-Conditional IIMA Factor Premia

Regime	Factor	n	Ann. Return %	Ann. Vol %	Sharpe	t-stat	p-value	Sig.
Bull	MKT	593	23.59	15.89	1.485	2.28	0.023	*
Bull	SMB	593	0.61	11.70	0.052	0.08	0.936	
Bull	HML	593	9.80	13.75	0.712	1.09	0.275	
Bull	WML	593	10.60	12.44	0.852	1.31	0.192	
Sideways	MKT	955	1.65	12.14	0.136	0.27	0.791	
Sideways	SMB	955	3.10	11.15	0.278	0.54	0.588	
Sideways	HML	955	6.15	11.58	0.531	1.03	0.302	
Sideways	WML	955	20.77	12.46	1.667	3.24	0.001	**
Bear	MKT	121	37.55	37.43	1.003	0.70	0.488	
Bear	SMB	121	−32.91	21.66	−1.520	−1.05	0.294	
Bear	HML	121	45.29	20.07	2.256	1.56	0.121	
Bear	WML	121	−12.89	24.06	−0.536	−0.37	0.711	

Conditioning on HMM regime reveals substantial heterogeneity suppressed by full-sample averages.

Bull regime (n = 593). The market premium concentrates in Bull: MKT earns 23.59% annualised (SR = 1.485, $t = 2.28^*$). Momentum continues to earn (WML: 10.60%, SR = 0.852) but is not statistically significant within regime. Value (HML) and size (SMB) premia are economically moderate and insignificant.

Sideways regime (n = 955). The momentum premium dominates: WML earns 20.77% annualised (SR = 1.667, t = 3.24) — **the strongest and most statistically significant regime-factor result in our analysis.**¹ **This finding is economically intuitive: momentum strategies rely on trend persistence without reversal, which characterises directionless but trending markets. The result survives exclusion of the COVID-19 window (estimated SR = 1.54, t = 2.98).** MKT is near zero (1.65%) and insignificant in Sideways, consistent with the absence of a broad directional trend.

¹ *Note on identification.* The HMM emission variables include Nifty 50 20-day time-series momentum (the index’s own trailing return), used to distinguish Sideways from Bull/Bear. WML is the IIMA cross-sectional long-short momentum factor, constructed from individual stock relative returns. These are distinct objects: the emission indexes the market’s direction; WML captures dispersion in stock-level return continuation. Regime labeling is driven primarily by VIX level and the market return component, not WML return itself. The WML-Sideways result is therefore not a mechanical artifact of the regime definition.

Bear regime (n = 121). Statistical inference is limited by the sparse Bear sample; no factor achieves significance within regime. Directionally, HML is positive (45.29% annualised, SR = 2.256), consistent with defensive institutional rotation toward cheap, low-valuation stocks during drawdowns. SMB is directionally negative (−32.91%, SR = −1.520), consistent with flight from small-capitalisation names to liquid large-caps. MKT (37.55%, SR = 1.003) is positive but driven by volatile recovery days within the Bear window; the Bear classification captures high-VIX periods including both crash and early-recovery phases. WML is directionally negative (−12.89%, SR = −0.536), consistent with momentum crashes during market dislocations documented by Daniel and Moskowitz (2016).

Summary. The largest regime-conditional Sharpe ratio spread across all factor-regime pairs is for WML (spread = 2.20, Sideways high vs. Bear low) and MKT (spread = 1.35, Bull high vs. Sideways low). Factor premia are not stationary over the business cycle — they concentrate in specific regime states in a pattern that is both statistically detectable and economically interpretable.

5. FII Flows and Factor Premia: Granger Causality

5.1 Framework

We test Granger causality (Granger, 1969) in bivariate VAR systems. For each factor F and regime R , we estimate:

$$F_t = \alpha + \sum \beta_k F_{t-k} + \sum \gamma_k FII_{t-k} + \varepsilon_t$$

restricted to observations within regime R . The null $H_0: \gamma_1 = \dots = \gamma_p = 0$. We use SSR F-tests with Newey-West HAC standard errors (bandwidth = 5) to account for autocorrelation and heteroskedasticity (Newey and West, 1987). Lag order p is AIC-selected within each pair ($p_{\max} = 5$). Minimum 80 observations required per regime-factor cell.

Factor series for Granger tests. Section 4 uses IIMA’s four daily factors (MKT, SMB, HML, WML) as the primary regime-conditional premia analysis. Section 5 Panel B uses own-constructed daily factors — MKT, SMB, HML, RMW, and CMA — because the Granger framework requires the same daily frequency and the IIMA library does not include profitability (RMW) or investment (CMA). Cross-validation confirms own-constructed MKT has $r = 0.907$ with IIMA MKT, so the net-flow null for the market factor is robust to factor construction. SMB and HML Granger results from own-constructed factors should be interpreted with the construction caveat from Section 2.2 (near-zero IIMA correlation); we treat them as directional only and do not make strong claims from Bear×SMB or Bear×HML cells.

Multiple testing. The Panel B grid is 15 simultaneous tests (5 factors × 3 regimes). We apply Benjamini-Hochberg FDR correction (Benjamini and Hochberg, 1995) jointly at $q = 0.05$. Only FDR-surviving cells constitute confirmed findings.

Methodological pre-commitments. To address standard referee concerns upfront: (i) HMM look-ahead bias is eliminated by filtered forward-pass probabilities and t-1 regime conditioning; (ii) multiple testing inflation is controlled by BH-FDR across all 15 Panel B cells; (iii) regime-conditional factor premia (Section 4) use IIMA survivorship-adjusted factors as primary; (iv) COVID dominance is tested by replicating the panels excluding February–June 2020, subject to the Bear-sample constraint detailed in Section 8; (v) autocorrelation and heteroskedasticity in daily returns are corrected by Newey-West HAC standard errors throughout; (vi) mechanism claims are framed as “consistent with” rather than causal attribution, as aggregate flow data does not support stock-level inference.

5.2 Panel A — FII Net Flow vs. Aggregate Market Return

Table 3, Panel A — FII Net Flow → MKT Granger Causality

Sample	F-stat	p-value	Best lag	n
Pooled	1.07	0.300	1	1,669
Bull	0.47	0.491	1	615
Sideways	1.00	0.367	2	933
Bear	1.43	0.234	1	120

FII net flow does not Granger-cause aggregate MKT returns in any regime or in the pooled sample. This null eliminates the possibility that Panel B results reflect a broad market-level flow relationship.

5.3 Panel B — FII Net Flow vs. Factor Premia (3 regimes × 5 own-constructed factors)

Table 3, Panel B — FII → Factor Granger Causality Matrix

Regime	Factor	F-stat	p (raw)	p (BH-FDR)	Best lag	n	FDR reject
Bull	MKT	0.47	0.491	0.567	1	615	No
Bull	SMB	2.65	0.071	0.212	2	615	No
Bull	HML	2.06	0.084	0.212	4	615	No
Bull	RMW	1.51	0.222	0.390	2	615	No
Bull	CMA	1.98	0.080	0.212	5	615	No
Sideways	MKT	1.00	0.367	0.501	2	933	No
Sideways	SMB	2.06	0.085	0.212	4	933	No
Sideways	HML	0.75	0.556	0.595	4	933	No
Sideways	RMW	0.89	0.409	0.512	2	933	No
Sideways	CMA	1.21	0.305	0.458	3	933	No
Bear	MKT	1.43	0.234	0.390	1	120	No
Bear	SMB	8.21	0.005	0.071	1	120	No
Bear	HML	0.51	0.726	0.726	4	120	No
Bear	RMW	1.73	0.134	0.287	5	120	No
Bear	CMA	6.98	0.009	0.071	1	120	No

Across 15 tests, Bear × SMB ($F = 8.21$, $p_{\text{raw}} = 0.005$) and Bear × CMA ($F = 6.98$, $p_{\text{raw}} = 0.009$) exhibit the strongest raw signals. After Benjamini-Hochberg correc-

tion, neither survives (adjusted $p = 0.071$ for both; BH threshold for the rank-1 cell: $0.05 \times 1/15 = 0.0033$). All 15 cells are null after FDR adjustment.

This complete null is a substantive finding. The large regime-conditional factor patterns documented in Section 4 are not FII-flow driven. If foreign institutional trading drove regime-conditional premia, FII flows would Granger-cause the corresponding factor returns. The data reject this hypothesis across all 15 tests. The mechanism behind regime-conditional factor dynamics in India operates through channels beyond daily aggregate FII net investment — most likely domestic participation structure, liquidity segmentation, and macroeconomic regime shifts that affect all market participants simultaneously.

5.4 Panel C — FII as Bear Regime Early Warning

Identification note. One identification concern applies to any test that predicts HMM-defined regime states using FII flows: the HMM uses India VIX as an emission variable, and FII flows may be contemporaneously correlated with VIX (as foreign investors respond to implied volatility). Predicting HMM Bear with flows could therefore be partially circular — VIX drives Bear labels and VIX drives FII behavior simultaneously. We address this in two ways. First, we define Bear onset as the HMM transition filtered at $t-1$ (see Section 3), so the 5-day ahead prediction window is strictly forward-looking. Second, a robustness check using an objective drawdown criterion (Nifty 50 trailing 20-day drawdown $> 8\%$) as the Bear onset definition yields qualitatively identical results.

A logistic model using lagged FII flows (lags 1–5) to predict Bear regime onset within a 5-day forward window produces $\text{AUC-ROC} = 0.548$ and $\text{McFadden } R^2 = 0.004$. No individual lag coefficient is statistically significant. FII flows do not function as an early warning system for Bear onset at the 1–5 day horizon. This null is robust to the identification concern above: even if the regime definition were partially correlated with FII behavior, the model produces near-chance AUC, suggesting no predictive content exists. Panel C complements Panel B: FII flows neither predict factor-level premia (Panel B) nor signal imminent regime transitions (Panel C).

6. Buy-Sell Decomposition: Gross Flow as Bear-Regime Price Impact

6.1 Motivation

Net FII flow aggregates gross purchases and sales, potentially discarding information. We separately test gross purchases and gross sales as predictors of MKT returns within each regime.

6.2 Results

Table 3, Panel D — Gross Buy vs. Gross Sell → MKT by Regime

Regime	Gross purchases F (p)	Gross sales F (p)	Lag
Bull	— (p > 0.16)	— (p > 0.16)	—
Sideways	— (p > 0.05)	— (p > 0.05)	—
Bear	5.06 (0.008)	4.89 (0.009)	2
Pooled	— (p > 0.10)	— (p > 0.10)	—

In Bear regimes, both gross FII purchases ($F = 5.06$, $p = 0.008$, $\text{lag} = 2$) and gross FII sales ($F = 4.89$, $p = 0.009$, $\text{lag} = 2$) Granger-cause aggregate market returns. No such predictability exists in Bull or Sideways. The pooled full-sample tests are insignificant, explaining why net-flow studies miss this result.

Three aspects of this finding deserve emphasis.

First, the predictability is Bear-specific. In calm regimes, domestic participation absorbs FII order flow without systematic price impact. In Bear, domestic liquidity withdrawal concentrates price discovery in the FII segment. Large FII trades move prices and the impact persists for two trading days — consistent with evidence on institutional price impact during dislocations (Brunnermeier and Pedersen, 2009).

Second, the symmetry between gross purchase and gross sale predictability argues against information asymmetry (Myers and Majluf, 1984) as the mechanism. Informed exits would produce sell-dominant predictability; instead, both sides carry equal predictive content. The informative signal is FII trading volume — not direction. This is consistent with FII acting as the marginal price setter under stress rather than as an informed trader.

Third, this finding would be invisible using net flow data alone. Net flow cancels the gross components; by aggregating, the volume signal in Panel D is averaged away. Our gross flow decomposition — using NSDL’s published purchase/sale breakdown — is what makes this result detectable.

One caveat qualifies this finding. The Bear regime sample ($n = 121$) is dominated by the COVID-19 crisis window; excluding that window leaves only 46 Bear observations — below our minimum for reliable Granger inference (Section 8). The Bear gross-flow result is therefore not separable from the COVID episode: it is consistent with FII price impact under liquidity stress generally, but on present data we cannot rule out that it is specific to the 2020 crisis. Additional Bear episodes are required to distinguish these interpretations.

7. Regime-Aware Factor Allocation Strategy

A walk-forward strategy allocating to MVO (Max-Sharpe) weights in Bull, HRP weights (Lopez de Prado, 2018) in Sideways, and equal weights in Bear achieves the following performance over 2021–2026 (Table 4):

Table 4 — Strategy Comparison (walk-forward, 2021-2026, costs = 0.25% one-way per monthly rebalance)

Strategy	Total Ret %	Ann Ret %	Ann Vol %	Sharpe	Sortino	MaxDD %	Calmar
Regime-Aware	65.7	10.75	13.54	0.794	1.103	−17.5	0.615
Buy-and-Hold	57.7	9.75	13.31	0.733	1.026	−18.0	0.543
Static HRP	53.4	9.15	12.81	0.714	1.046	−18.6	0.492
MVO MaxSharpe	102.4	14.78	14.52	1.018	1.418	−15.0	0.988

The regime-aware strategy improves the Sharpe ratio by 0.061 over buy-and-hold (an 8.3% relative improvement) and reduces maximum drawdown by 0.5 percentage points. Neither strategy’s mean return is individually significant at conventional levels ($t = 1.80$ and 1.66 respectively), reflecting the short evaluation window. Notably, the static MVO Max-Sharpe strategy dominates all alternatives over this period (Sharpe 1.018) — suggesting that, conditional on the 2021–2026 sample, mean-variance optimization benefits from the specific return structure more than the regime signal adds. We report this transparently: the regime identification documented in Sections 3–4 is statistically real, but converting it into allocation alpha net of costs over a short sample is not established by these results.

Regime-aware turnover (317% annualised) is high and represents a practical constraint for strategies with significant AUM. This finding motivates lower-frequency regime conditioning (monthly rather than daily), which we consider an extension of the current work.

8. Robustness

COVID exclusion. We replicate the regime-conditional factor tests excluding the COVID-19 crisis window (2020-02-01 to 2020-06-30). The WML-Sideways result survives ($SR = 1.54$, $t = 2.98^{**}$). For the Panel B Granger grid, excluding the crisis window leaves only 46 Bear observations — below our 80-observation minimum — so the five Bear cells cannot be tested ex-COVID; this caveat applies equally to the Panel D Bear gross-flow result, which is therefore not separable from the COVID episode and should be read accordingly. Among the 10 testable (Bull and Sideways) cells, one cell — Bull $\times$ CMA ($F = 2.66$, $p_{\text{raw}} = 0.022$) — is significant at the 5% level before multiple-testing correction but does not survive Benjamini-Hochberg FDR at $q = 0.05$ (rank-1 threshold 0.005); all other cells are null at raw and adjusted levels. The ex-COVID Panel B grid therefore contains no confirmed findings, consistent with the full-sample null.

Alternative factor data. Appendix A2 reports regime-conditional results using own-constructed daily factors (MKT, SMB, HML, RMW, CMA) from the 14-stock NSE blue-chip universe. MKT results are consistent with IIMA ($r = 0.907$ between series). SMB and HML diverge substantially ($r = -0.18$ and 0.01 respectively), attributable to the absence of true small-cap and value-dispersion in the blue-chip universe. We use IIMA as primary and flag own-constructed SMB/HML results as unreliable in Appendix A2.

HAC standard errors. All F-statistics throughout use Newey-West HAC covariance (bandwidth = 5). Full-sample results assuming homoskedastic errors are reported in Appendix A4 for comparison; the correction is material in several high-autocorrelation factor series.

Alternative regime definition. As a robustness check, we define Bull (Bear) as periods where the 63-day rolling Nifty return exceeds (falls below) +10% (−10%), with Sideways as the remainder. The WML-Sideways direction persists; the Bear sample under this definition is $n = 89$, further limiting inference on Bear-specific results.

9. Limitations

Short sample and Bear sparsity. Our primary sample spans seven calendar years (2019–2025). The Bear regime covers only 121 trading days, concentrated in the

COVID-19 crash (2020) and the 2022 global tightening episode. Bear-regime results lack statistical power; effect size estimates (Sharpe ratios) are unstable and should be interpreted directionally, not as precise parameter estimates.

Factor construction. IIMA factors use point-in-time accounting data and a broad BSE/NSE universe with micro-cap filters. Our own-constructed factors use yfinance fundamentals and a 14-stock NSE blue-chip universe, producing near-zero SMB and HML correlation with IIMA equivalents. This divergence underscores the importance of the robustness check and the hazard of constructing size/value factors from a restricted large-cap universe. IIMA does not provide RMW or CMA; we cannot benchmark the profitability and investment factors against a canonical Indian source.

Aggregate flow data. We observe total daily FII net equity investment, not individual fund or stock-level flows. Attribution of the Bear gross-flow predictability to specific investment behaviors (momentum rebalancing, risk parity deleveraging, mandate-driven redemptions) is interpretive. Verification using quarterly NSDL FPI stock-level holdings data is a natural extension.

Survivorship bias. Our own-constructed NSE universe excludes stocks delisted over 2019–2026 (YES Bank restructuring, IL&FS subsidiaries). This bias inflates value and profitability premia directionally, making our null results for these factors conservative and any positive findings potentially overstated.

DCC as concurrent indicator. The DCC average pairwise correlation rises significantly in Bear regimes (mean 0.30 vs. 0.25 in Bull), confirming realized diversification failure. However, DCC is a concurrent rather than leading indicator — India VIX Granger-causes DCC but not vice versa, indicating options markets price stress before realized correlation reflects it. Practitioners relying on realized correlation as a real-time risk trigger face a systematic lag relative to implied volatility signals.

Regime-strategy turnover. The regime-aware strategy generates 317% annualised turnover at monthly rebalancing. In live implementation, this constrains scalable AUM and requires execution infrastructure that the current framework does not model.

10. Conclusion

We examine factor premia dynamics across HMM-identified market regimes in Indian equities and the role of FII flows in explaining them.

Three findings emerge. First, factor premia are regime-conditional using the canon-

ical IIM Ahmedabad factor library. The momentum premium (WML) concentrates in Sideways regimes (Sharpe 1.667, $t = 3.24^{**}$) — the strongest statistically significant regime-factor result in our analysis and the most actionable for practitioners with systematic momentum strategies. The market premium concentrates in Bull (Sharpe 1.485, $t = 2.28^*$). Value (HML) is directionally positive in Bear, consistent with defensive rotation into low-valuation stocks, but Bear sample size limits formal inference.

Second, aggregate FII net flows do not Granger-cause individual factor premia in any of 15 regime-factor tests after Benjamini-Hochberg FDR correction. This rules out daily foreign institutional trading as the mechanism behind the regime-conditional patterns documented above. We explicitly feature this null as a contribution: rigorously executed negative results are informative and reduce the risk of spurious factor-flow attribution that characterises much of the practitioner literature on foreign flows in India.

Third, a gross flow decomposition reveals that both FII purchases and sales Granger-cause aggregate market returns specifically in Bear regimes ($F \approx 5$, $p \approx 0.009$), with symmetry between buy and sell sides pointing to volume-driven price impact rather than information asymmetry. This Bear-specific market predictability is invisible when using net flow data alone — it requires the gross flow decomposition that NSDL data supports but most aggregate studies do not exploit. Because the Bear sample is dominated by the COVID-19 episode, this finding awaits confirmation in future Bear regimes before it can be read as a general feature of Indian market stress.

A regime-aware factor allocation strategy achieves Sharpe 0.794 versus 0.733 for buy-and-hold over a 2021–2026 walk-forward period with lower maximum drawdown, although a static mean-variance allocation outperforms both over this sample — converting regime identification into robust allocation alpha remains an open implementation question.

Future work should examine the FII-DII flow interaction channel — whether the growth in domestic SIP and DII flows post-2020 has structurally reduced Indian equities’ fragility to FII outflow shocks — and use quarterly NSDL FPI stock-level holdings to verify whether gross flow price impact operates through specific factor-bucket concentration.

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Appendix

A2: Own-Constructed Five-Factor Regime Results

Table A2 — Own-Constructed Factor Premia by Regime (14-stock NSE universe, 1,729 days)

Regime	Factor	n	Ann. Return %	Ann. Vol %	Sharpe	t-stat	p-value
Full	MKT	1,729	3.72	17.96	0.207	0.54	0.587
Full	SMB	1,729	2.21	12.70	0.174	0.46	0.649
Full	HML	1,729	3.34	12.28	0.272	0.71	0.476
Full	RMW	1,729	-3.11	12.47	-0.249	-0.65	0.514
Full	CMA	1,729	3.23	14.36	0.225	0.59	0.556
Bull	MKT	632	14.01	16.73	0.838	1.33	0.185
Bull	SMB	632	-1.43	12.65	-0.113	-0.18	0.858
Bull	HML	632	6.92	12.58	0.550	0.87	0.383
Bull	RMW	632	-8.89	12.37	-0.719	-1.14	0.255
Bull	CMA	632	6.62	14.77	0.448	0.71	0.478
Sideways	MKT	976	-3.72	11.17	-0.333	-0.65	0.513
Sideways	SMB	976	-1.04	10.03	-0.103	-0.20	0.839
Sideways	HML	976	4.64	10.02	0.463	0.91	0.362
Sideways	RMW	976	-1.97	10.05	-0.196	-0.39	0.699
Sideways	CMA	976	4.81	11.43	0.421	0.83	0.407
Bear	MKT	121	10.01	46.44	0.216	0.15	0.881
Bear	SMB	121	47.37	25.56	1.853	1.28	0.199
Bear	HML	121	-25.89	22.82	-1.135	-0.79	0.432
Bear	RMW	121	17.97	24.73	0.727	0.50	0.615
Bear	CMA	121	-27.25	27.53	-0.989	-0.69	0.493

Own-constructed MKT has $r = 0.907$ correlation with IIMA MKT (high confidence). SMB ($r = -0.18$) and HML ($r = 0.01$) diverge substantially from IIMA equivalents due to absence of true size/value dispersion in the 14-stock blue-chip universe. SMB

Bear (SR = +1.853) and HML Bear (SR = -1.135) results should not be relied upon, as this construction cannot identify the size or value premium reliably. RMW and CMA results are reported for completeness; no IIMA benchmark is available.

A3: FII Granger Causality — Full Matrix

Reported in Table 3, Panel B (main text). Raw p-values and BH-FDR adjusted p-values for all 15 regime-factor cells. Lag selection by AIC ($p_{\max} = 5$). HAC standard errors throughout. Replication output: `outputs/table3_panel_b.csv`.

A4: COVID-Exclusion Robustness

All Granger tests replicated excluding 2020-02-01 to 2020-06-30. WML-Sideways: SR = 1.54, $t = 2.98^{**}$. Panel B: five Bear cells untestable ex-COVID ($n = 46 < 80$ -observation minimum); of the 10 testable Bull/Sideways cells, Bull $\times$ CMA is raw-significant ($F = 2.66$, $p = 0.022$) but fails BH-FDR at $q = 0.05$; all other cells null. Panel D Bear gross-flow: untestable ex-COVID for the same sample reason and therefore not separable from the crisis episode. Replication outputs: `outputs/table3_panel_b_covid_robustness.csv` and companion panels.

Code, data, and replication scripts: github.com/YASHpPATIL7/alpha-core (paper_analysis/)

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